

System Design Document

Heart Disease Prediction System

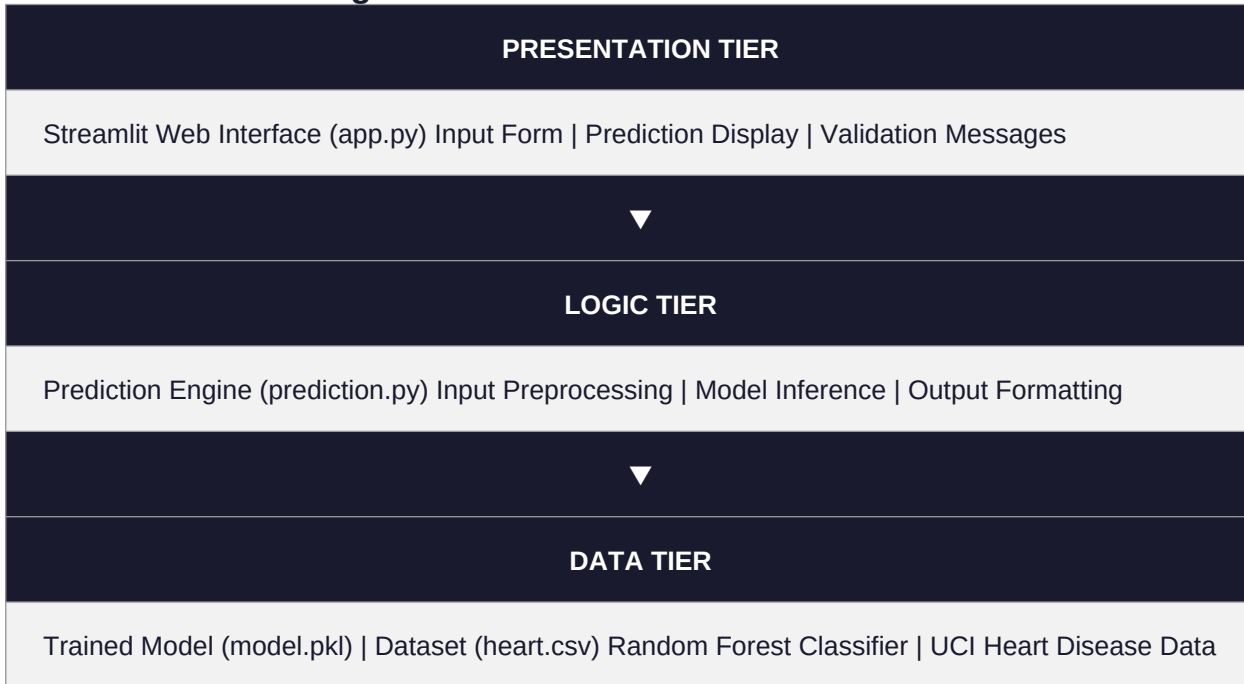
Version 1.0 | 2025

Covers: Architecture, Modules, Data Flow, Database Design

1. System Architecture Overview

The Heart Disease Prediction System follows a three-tier architecture: Presentation Tier (Streamlit UI), Logic Tier (Prediction Engine), and Data Tier (Dataset + Serialised Model). All tiers reside within a single Python application, making deployment straightforward.

1.1 Architecture Diagram



2. Module Design

2.1 app.py — Streamlit Application

- Renders the patient input form with 13 feature fields
- Calls prediction.py on form submission
- Displays prediction result and confidence percentage

- Handles input validation feedback

2.2 train_model.py — Model Training

- Loads and pre-processes heart.csv
- Splits data into training (80%) and testing (20%) sets
- Trains a Random Forest Classifier
- Evaluates model (accuracy, confusion matrix, classification report)
- Serialises the model to model.pkl via Joblib

2.3 prediction.py — Prediction Engine

- Loads model.pkl at module import time
- Accepts a dictionary of 13 patient features
- Converts inputs to a NumPy array in the correct feature order
- Returns predicted class (0/1) and probability scores

3. Data Flow Diagram

Step	Actor / Process	Data
1	User	Enters 13 patient attributes into the web form
2	Streamlit (app.py)	Validates inputs; raises error if invalid
3	app.py → prediction.py	Passes validated feature dictionary
4	prediction.py	Loads model.pkl and preprocesses features
5	Random Forest Model	Computes prediction class and probability
6	prediction.py → app.py	Returns (prediction, probability) tuple
7	Streamlit (app.py)	Displays result to user on screen

4. Machine Learning Design

4.1 Algorithm Selection

Algorithm	Accuracy (Approx.)	Selected?
Logistic Regression	82%	No
Decision Tree	79%	No
K-Nearest Neighbours	81%	No
Random Forest	87%	Yes ✓
SVM	84%	No

4.2 Training Pipeline

1. Load dataset → `pandas.read_csv('dataset/heart.csv')`
2. Feature / Target → `X = df.drop('target', axis=1)`
`y = df['target']`
3. Train-Test Split → `train_test_split(X, y, test_size=0.2, random_state=42)`
4. Model Training → `RandomForestClassifier(n_estimators=100, random_state=42)`
5. Model Evaluation → `accuracy_score, confusion_matrix, classification_report`
6. Save Model → `joblib.dump(model, 'model.pkl')`

4.3 Feature Importance

Rank	Feature	Importance (Approx.)
1	thalach (Max Heart Rate)	High
2	cp (Chest Pain Type)	High
3	ca (Major Vessels)	High

Rank	Feature	Importance (Approx.)
4	thal (Thalassemia)	Medium
5	oldpeak (ST Depression)	Medium
6	age	Medium
7	exang (Exercise Angina)	Medium

5. Interface Design

5.1 Input Form Fields

Field	Input Type	Valid Range / Options
Age	Number	1 – 120
Sex	Selectbox	Male / Female
Chest Pain Type	Selectbox	0, 1, 2, 3
Resting BP	Number	60 – 200 mm Hg
Cholesterol	Number	100 – 600 mg/dl
Fasting Blood Sugar	Selectbox	0 / 1
Resting ECG	Selectbox	0, 1, 2
Max Heart Rate	Number	60 – 220 bpm
Exercise Angina	Selectbox	0 / 1
ST Depression	Number (float)	0.0 – 10.0
Slope	Selectbox	0, 1, 2
CA	Selectbox	0 – 4
Thal	Selectbox	0, 1, 2, 3